Collect Inference Metrics from Triton Inference Server

NetApp Solutions

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February 18, 2021

This PDF was generated from https://docs.netapp.com/us-en/netapp-solutions/ai/hciai_edge_collect_inference_metrics_from_triton_inference_server.html on July 11, 2021.
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Collect Inference Metrics from Triton Inference Server

The Triton Inference Server provides Prometheus metrics indicating GPU and request statistics.

By default, these metrics are available at "http://<triton_inference_server_IP>:8002/metrics".

The Triton Inference Server IP is the LoadBalancer IP that was recorded earlier.

The metrics are only available by accessing the endpoint and are not pushed or published to any remote server.

172.21.231.132:8002/metrics

Not secure | 172.21.231.132:8002/metrics

```
# HELP nv_inference_request_success Number of successful inference requests, all batch sizes
# TYPE nv_inference_request_success counter
nv_inference_request_success{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1"} 6.000000
nv_inference_request_success{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1"} 4.000000
nv_inference_request_success{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1"} 5.000000
# HELP nv_inference_request_failure Number of failed inference requests, all batch sizes
# TYPE nv_inference_request_failure counter
# HELP nv_inference_count Number of inferences performed
# TYPE nv_inference_count counter
nv_inference_count{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1"} 260.000000
nv_inference_count{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1"} 4.000000
nv_inference_count{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1"} 5.000000
# HELP nv_inference_exec_count Number of model executions performed
# TYPE nv_inference_exec_count counter
nv_inference_exec_count{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1"} 6.000000
nv_inference_exec_count{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1"} 4.000000
nv_inference_exec_count{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1"} 5.000000
# HELP nv_inference_request_duration_us Cummulative inference request duration in microseconds
# TYPE nv_inference_request_duration_us counter
nv_inference_request_duration_us{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1"} 2172236.000000
nv_inference_request_duration_us{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1"} 1042062.000000
nv_inference_request_duration_us{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1"} 1476198.000000
# HELP nv_inference_compute_duration_us Cummulative inference compute duration in microseconds
# TYPE nv_inference_compute_duration_us counter
nv_inference_compute_duration_us{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1"} 2159478.000000
nv_inference_compute_duration_us{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1"} 1041291.000000
nv_inference_compute_duration_us{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1"} 1475336.000000
# HELP nv_inference_queue_duration_us Cummulative inference queuing duration in microseconds
# TYPE nv_inference_queue_duration_us counter
nv_inference_queue_duration_us{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1"} 514.000000
nv_inference_queue_duration_us{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1"} 378.000000
nv_inference_queue_duration_us{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1"} 366.000000
# TYPE nv_inference_load_ratio histogram
nv_inference_load_ratio_count{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1"} 6
nv_inference_load_ratio_sum{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1"} 6.053677
nv_inference_load_ratio_bucket{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1",le="1.050000"} 6
nv_inference_load_ratio_bucket{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1",le="1.100000"} 6
nv_inference_load_ratio_bucket{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1",le="1.250000"} 6
nv_inference_load_ratio_bucket{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1",le="1.500000"} 6
nv_inference_load_ratio_bucket{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1",le="2.000000"} 6
nv_inference_load_ratio_bucket{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1",le="10.000000"} 6
nv_inference_load_ratio_bucket{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1",le="50.000000"} 6
nv_inference_load_ratio_bucket{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f",model="resnet50_netdef",version="1",le="+Inf"} 6
nv_inference_load_ratio_count{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1"} 4
nv_inference_load_ratio_sum{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1"} 4.032081
nv_inference_load_ratio_bucket{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1",le="1.050000"} 4
nv_inference_load_ratio_bucket{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1",le="1.100000"} 4
nv_inference_load_ratio_bucket{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1",le="1.250000"} 4
nv_inference_load_ratio_bucket{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1",le="1.500000"} 4
nv_inference_load_ratio_bucket{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1",le="2.000000"} 4
nv_inference_load_ratio_bucket{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1",le="10.000000"} 4
nv_inference_load_ratio_bucket{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1",le="50.000000"} 4
nv_inference_load_ratio_bucket{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958",model="resnet50_netdef",version="1",le="+Inf"} 4
nv_inference_load_ratio_count{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1"} 5
nv_inference_load_ratio_sum{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1"} 5.033626
nv_inference_load_ratio_bucket{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1",le="1.050000"} 5
nv_inference_load_ratio_bucket{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1",le="1.100000"} 5
nv_inference_load_ratio_bucket{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1",le="1.250000"} 5
nv_inference_load_ratio_bucket{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1",le="1.500000"} 5
nv_inference_load_ratio_bucket{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1",le="2.000000"} 5
nv_inference_load_ratio_bucket{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1",le="10.000000"} 5
nv_inference_load_ratio_bucket{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1",le="50.000000"} 5
nv_inference_load_ratio_bucket{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1",le="+Inf"} 5
```

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nv_inference_load_ratio_bucket{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1",model="resnet50_netdef",version="1",le="+Inf"} 5
# HELP nv_gpu_utilization GPU utilization rate [0.0 - 1.0)
# TYPE nv_gpu_utilization gauge
nv_gpu_utilization{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1"} 0.000000
nv_gpu_utilization{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f"} 0.000000
nv_gpu_utilization{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958"} 0.000000
# HELP nv_gpu_memory_total_bytes GPU total memory, in bytes
# TYPE nv_gpu_memory_total_bytes gauge
nv_gpu_memory_total_bytes{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1"} 15843721216.000000
nv_gpu_memory_total_bytes{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f"} 15843721216.000000
nv_gpu_memory_total_bytes{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958"} 15843721216.000000
# HELP nv_gpu_memory_used_bytes GPU used memory, in bytes
# TYPE nv_gpu_memory_used_bytes gauge
nv_gpu_memory_used_bytes{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1"} 1466236928.000000
nv_gpu_memory_used_bytes{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f"} 13004767232.000000
nv_gpu_memory_used_bytes{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958"} 1466236928.000000
# HELP nv_gpu_power_usage GPU power usage in watts
# TYPE nv_gpu_power_usage gauge
nv_gpu_power_usage{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1"} 27.999000
nv_gpu_power_usage{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f"} 28.428000
nv_gpu_power_usage{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958"} 27.632000
# HELP nv_gpu_power_limit GPU power management limit in watts
# TYPE nv_gpu_power_limit gauge
nv_gpu_power_limit{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1"} 70.000000
nv_gpu_power_limit{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f"} 70.000000
nv_gpu_power_limit{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958"} 70.000000
# HELP nv_energy_consumption GPU energy consumption in joules since the Triton Server started
# TYPE nv_energy_consumption counter
nv_energy_consumption{gpu_uuid="GPU-b882076d-0b82-1b8b-5b05-9762986e8ee1"} 9796.449000
nv_energy_consumption{gpu_uuid="GPU-28a3f0dc-400f-e494-809c-f439ac1afc4f"} 9997.538000
nv_energy_consumption{gpu_uuid="GPU-aef8cff6-9325-0a1d-0937-ee91a4332958"} 9669.536000
```

Next: Validation Results

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